

Stochastic Processes, Long Quiz, 2026 Fall

- Duration: 60 minutes
 - Weight: 10% of final grade
 - Closed material, No calculator
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- Name: _____
 - Student ID: _____
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- Write legibly.
 - Justification is necessary unless stated otherwise.

#1. Let $X \sim U(0, 4)$. Evaluate $\mathbb{E}[\min(X, 3)]$. [2.5 pt]

#2. Let $X \sim \text{Poi}(3)$ and $Y = \min(X, 2)$. Find $\mathbb{E}[Y]$ and $\text{Var}(Y)$. [2.5 pt]

#3. Let $P = \begin{bmatrix} 0.75 & 0.25 \\ 0.50 & 0.50 \end{bmatrix}$.

(a) Find $\boldsymbol{\pi} = (\pi_1 \ \pi_2)$ satisfying $\boldsymbol{\pi}P = \boldsymbol{\pi}$ and $\pi_1 + \pi_2 = 1$. [2.5 pt]

(b) Find $\boldsymbol{\pi}P^2$. [2.5 pt]